Lines and Planes in Three Dimensions

Section 1: Vector and Cartesian Equations of Lines

1 Find the vector equation of the line passing through point $A(2,\ -1, 4)$ in the direction of $\mathbf{d} = [3, 5,\ -2]$.

a. $\mathbf{r} = [2,\ -1, 4] + \lambda[3, 5,\ -2]$

b. $\mathbf{r} = [3, 5,\ -2] + \lambda[2,\ -1, 4]$

c. $\mathbf{r} = [2 + 3\lambda,\ -1 - 5\lambda, 4 - 2\lambda]$

d. $\mathbf{r} = [5, 4, 2] + \lambda[1, 1, 1]$

2 Convert the symmetric Cartesian line equations $\frac{x-3}{2} = \frac{y+1}{-4} = \frac{z-5}{1}$ into a direction vector.

a. $[2,\ -4, 1]$

b. $[3,\ -1, 5]$

c. $[-2, 4,\ -1]$

d. $[1/2,\ -1/4, 1]$

3 Two lines in 3D that do not intersect and are not parallel are called:

a. Skew lines

b. Coplanar lines

c. Coincident lines

d. Asymptotic lines

Section 2: Vector and Cartesian Equations of Planes

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the Cartesian equation of the plane passing through $P(1, 2, -3)$ with normal vector $\mathbf{n} = [2, -4, 5]$.

a. $2x - 4y + 5z = -21$

b. $2x - 4y + 5z = 21$

c. $x + 2y - 3z = -21$

d. $2x + 4y + 5z = 0$

5 What is a normal vector to the plane $3x - 7y + z = 14$?

a. $[3, -7, 1]$

b. $[3, 7, 1]$

c. $[14, 14, 14]$

d. $[-3, 7, 14]$

6 Find a vector perpendicular to the plane containing the non-collinear vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\mathbf{u} \times \mathbf{v}$

b. $\mathbf{u} \cdot \mathbf{v}$

c. $\mathbf{u} + \mathbf{v}$

d. $\mathbf{u} - \mathbf{v}$

Section 3: Intersections and Angles

7 Find the intersection point of line $\mathbf{r} = [1 + 2t, -1 + t, 3 - t]$ with plane $x + 2y - z = 4$.

a. $t = 1 \Rightarrow (3, 0, 2)$

b. $t = 0 \Rightarrow (1, -1, 3)$

c. $t = 2 \Rightarrow (5, 1, 1)$

d. No intersection

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 How is the acute angle θ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ calculated?

a. $\sin\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

b. $\cos\theta = \frac{|\mathbf{d}\cdot\mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

c. $\tan\theta = \frac{|\mathbf{d}\times\mathbf{n}|}{|\mathbf{d}\cdot\mathbf{n}|}$

d. $\theta = \arccos(\mathbf{d}\cdot\mathbf{n})$

9 Find the acute angle between two planes with normal vectors $\mathbf{n}_1 = [1, 0, 1]$ and $\mathbf{n}_2 = [0, 1, 1]$.

a. $\frac{\pi}{3}$ (60°)

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{2}$ (90°)

d. $\frac{\pi}{6}$ (30°)

Section 4: Shortest Distances in 3D

10 What is the shortest distance from point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz = D$?

a. $\frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$

b. $\frac{|Ax_0 + By_0 + Cz_0 + D|}{A^2 + B^2 + C^2}$

c. $\sqrt{(x_0 - A)^2 + (y_0 - B)^2 + (z_0 - C)^2}$

d. $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$

11 Find the distance from the origin $(0, 0, 0)$ to the plane $2x - 3y + 6z = 28$.

a. 4

b. 28

c. $\frac{28}{7} = 4$

d. $\frac{28}{49}$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

12 How is the shortest distance between two parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 14$ calculated?

a. $\frac{|14 - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{9}{3} = 3$

b. $\frac{14 + 5}{3} = \frac{19}{3}$

c. $14 - 5 = 9$

d. $\frac{9}{9} = 1$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647